DVAPI Vulnerability Assessment Report

Report Date	April 14, 2026
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Target Application	DVAPI (Damn Vulnerable API)
Target URL	http://192.168.18.16:3000
Assessment Type	Black-box API Penetration Test
Classification	Confidential

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1. Executive Summary

A manual penetration test was conducted against DVAPI, a deliberately vulnerable API platform. Five distinct vulnerabilities were identified spanning the OWASP API Security Top 10 (2023). All findings were exploited successfully without brute force or automated scanning — through logic flaws and missing server-side controls alone.

The most critical finding (FIND-05) allowed any authenticated user to permanently delete other users' accounts via an unsecured HTTP method. FIND-02 and FIND-04 exposed private user data and allowed score manipulation through predictable parameter tampering. FIND-01 revealed a verbose error response leaking full server stack traces. FIND-03 exposed unreleased application content through hidden API parameters.

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2. Scope & Methodology

In Scope: `http://192.168.18.16:3000` — all API endpoints reachable through the DVAPI web interface.

Tools Used: Burp Suite (Proxy, Repeater, JWT Editor extension), browser (Chrome).

Methodology: Manual testing. All requests were intercepted via Burp Proxy, analyzed, and replayed in Repeater with modified parameters to verify authorization flaws and misconfiguration.

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3. Vulnerability Summary

ID	Vulnerability	Endpoint	Severity	OWASP Category
FIND-01	Security Misconfiguration — Stack Trace Disclosure	`POST /api/addNote`	High	API8:2023
FIND-02	Broken Object Level Authorization	`GET /api/getNote`	High	API1:2023
FIND-03	Improper Inventory Management	`POST /api/allChallenges`	Medium	API9:2023
FIND-04	Broken Object Property Level Authorization	`POST /api/register`	High	API3:2023
FIND-05	Broken Function Level Authorization	`DELETE /api/user/:username`	Critical	API5:2023

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4. Findings

FIND-01 — API8:2023 Security Misconfiguration

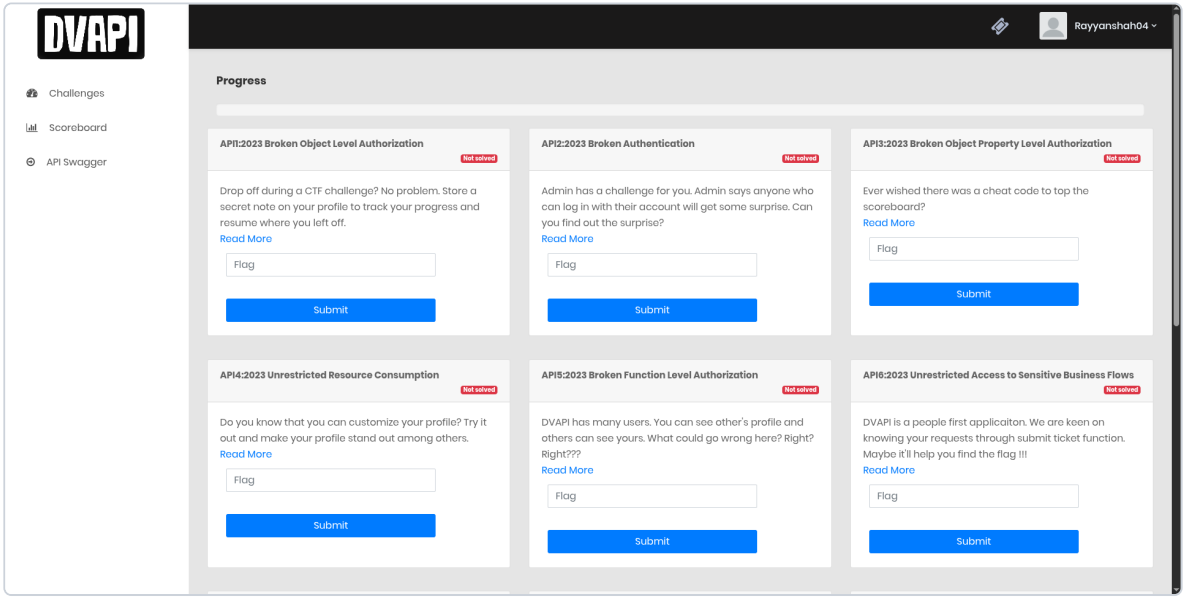
Severity	High
Endpoint	`POST /api/addNote`
Auth Required	Yes (JWT)
CVSS (approx.)	7.5

Description

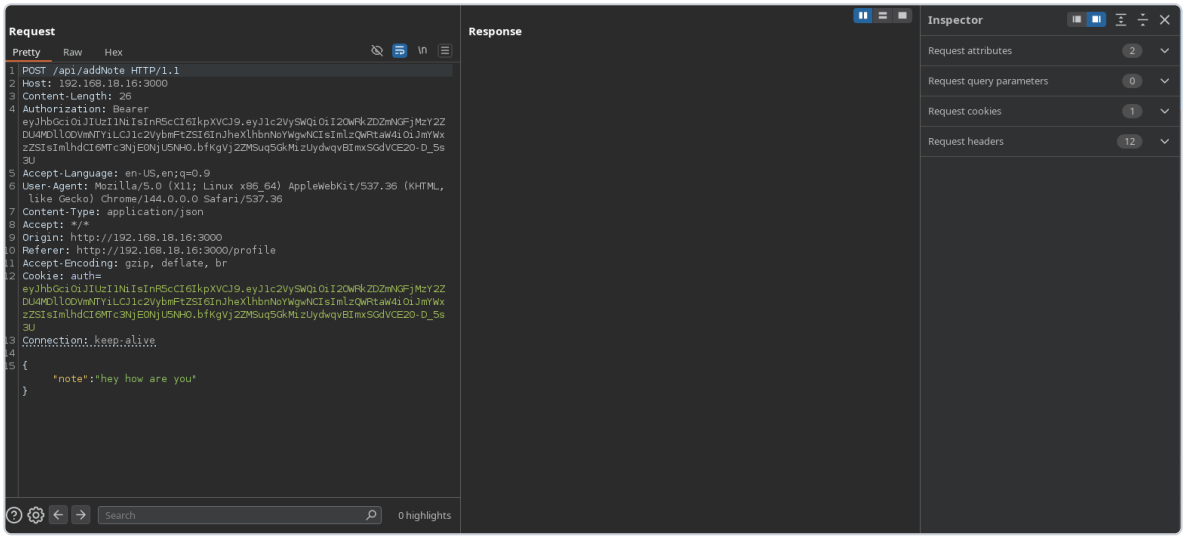
The API returns full Node.js stack traces in error responses instead of generic error messages. This discloses internal file paths, module names, framework version, and application logic to any authenticated user who triggers a validation error.

Steps to Reproduce

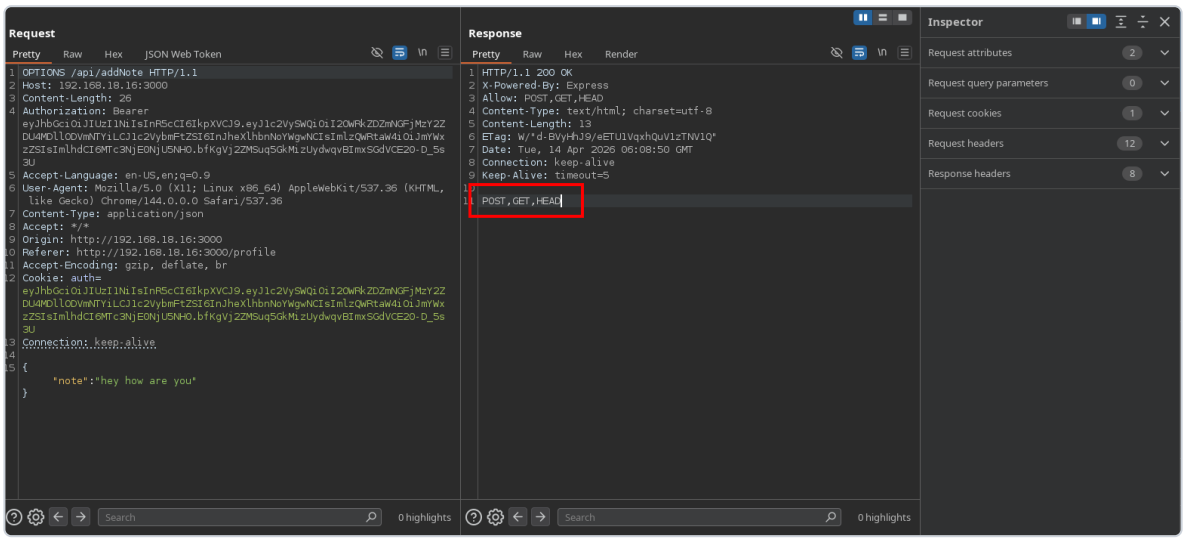
1. Log in and navigate to the profile page. Add a note to capture the `POST /api/addNote` request in Burp.



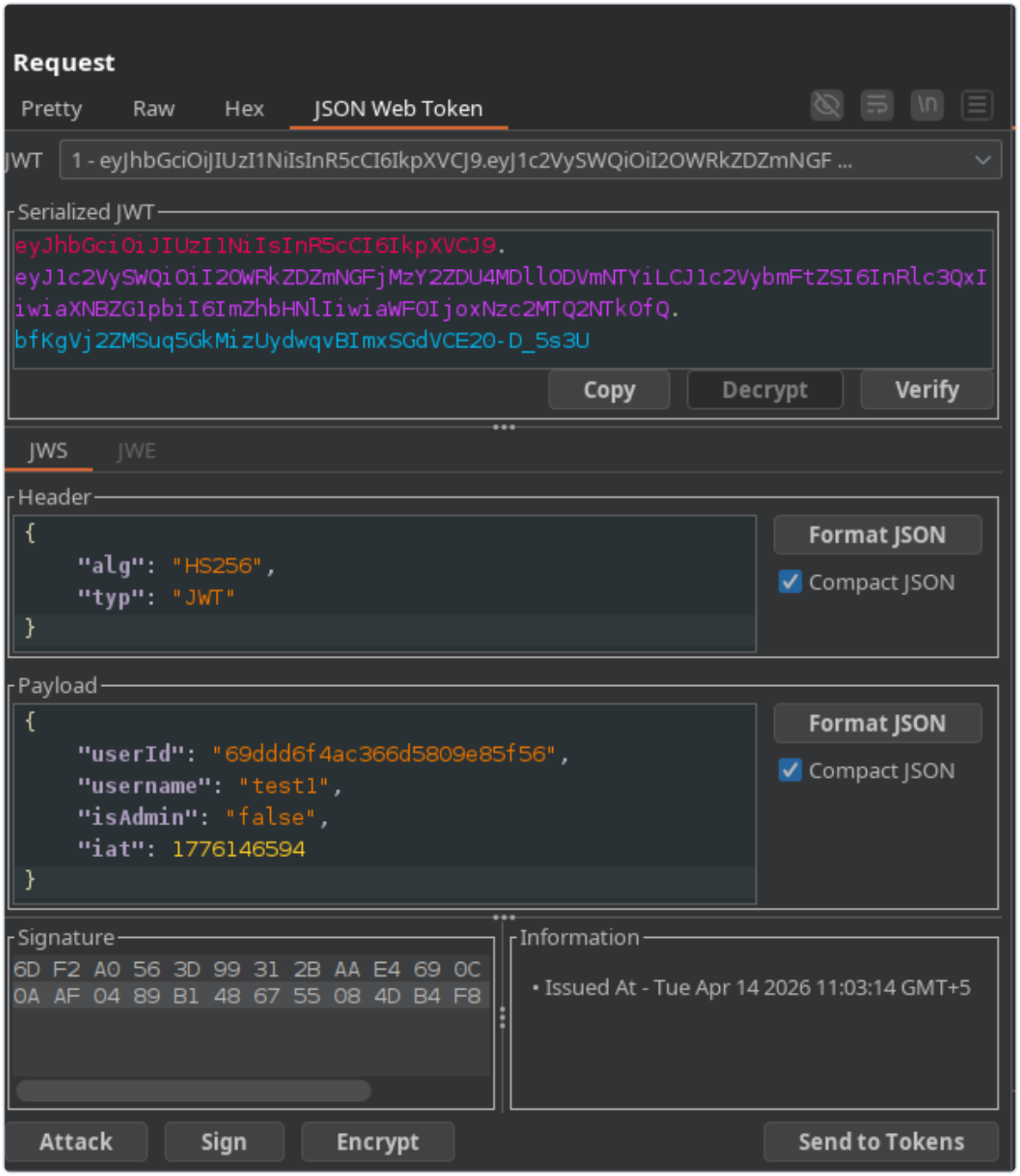
2. Send the captured request to Repeater. The request carries a JWT in both the `Authorization` header and `auth` cookie.



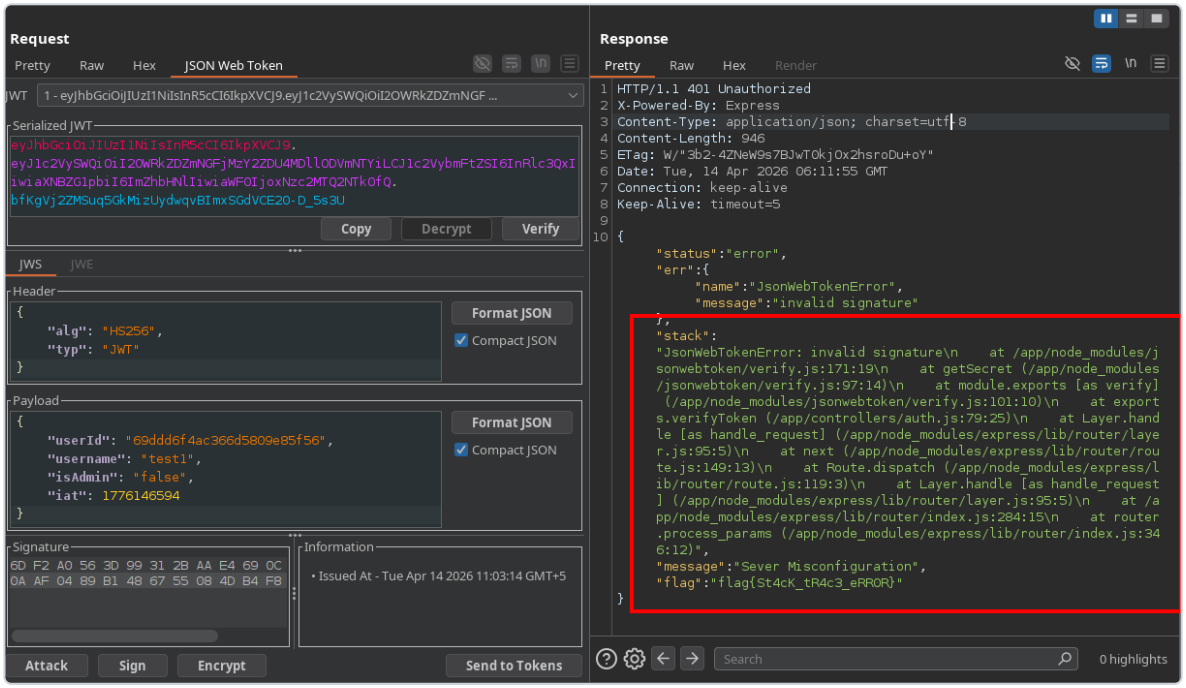
3. Use the OPTIONS method to enumerate what HTTP methods the endpoint accepts.



4. Using the JWT Editor extension, decode the current token. Observe it contains `userId`, `username`, and `isAdmin: false` claims, signed with HS256.



5. Modify the JWT payload (e.g., change `isAdmin` to `true`) and re-send. The server rejects the tampered signature with HTTP 401 — but the response body leaks a **full stack trace** along with the flag, confirming verbose error output is enabled in production.




```
Flag: flag{St4cK_tR4c3_eRR0R}
```

Impact

Attackers gain insight into server internals (file paths, library versions, code flow). This information is used to identify further attack vectors and aids in crafting targeted exploits.

Remediation

- Set `NODE_ENV=production` to suppress stack traces in Express.
- Implement a global error handler that returns only generic messages (`{"error": "Internal Server Error"}`).
- Never expose framework-level errors to API consumers.

FIND-02 — API1:2023 Broken Object Level Authorization (BOLA)

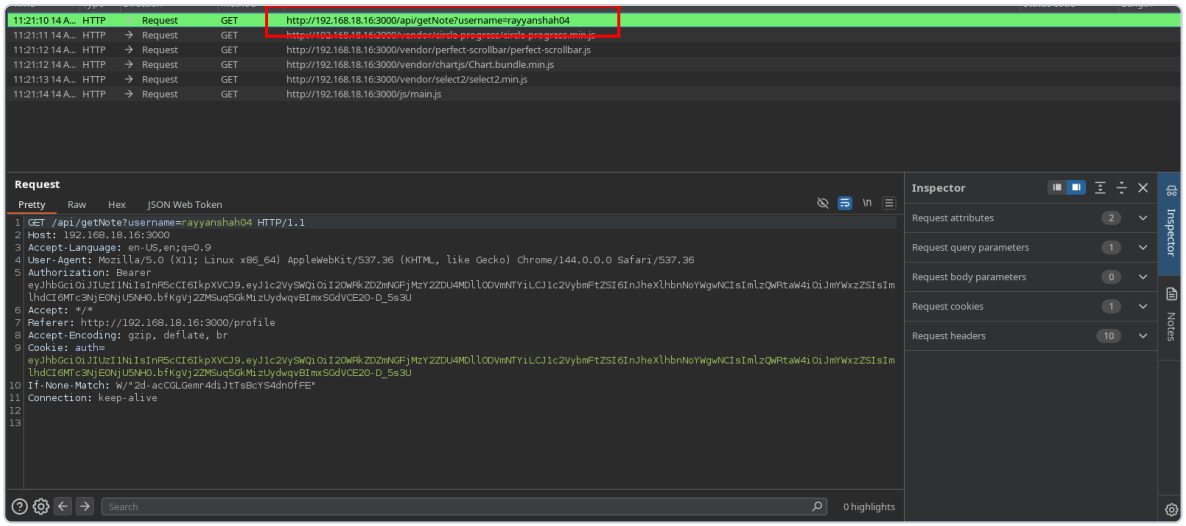
Severity	High
Endpoint	`GET /api/getNote?username=<target>`
Auth Required	Yes (JWT)
CVSS (approx.)	8.1

Description

The note retrieval endpoint accepts a `username` query parameter to identify whose note to return. The server does not verify that the requesting user owns the requested resource. Any authenticated user can read any other user's private note by simply supplying a different username.

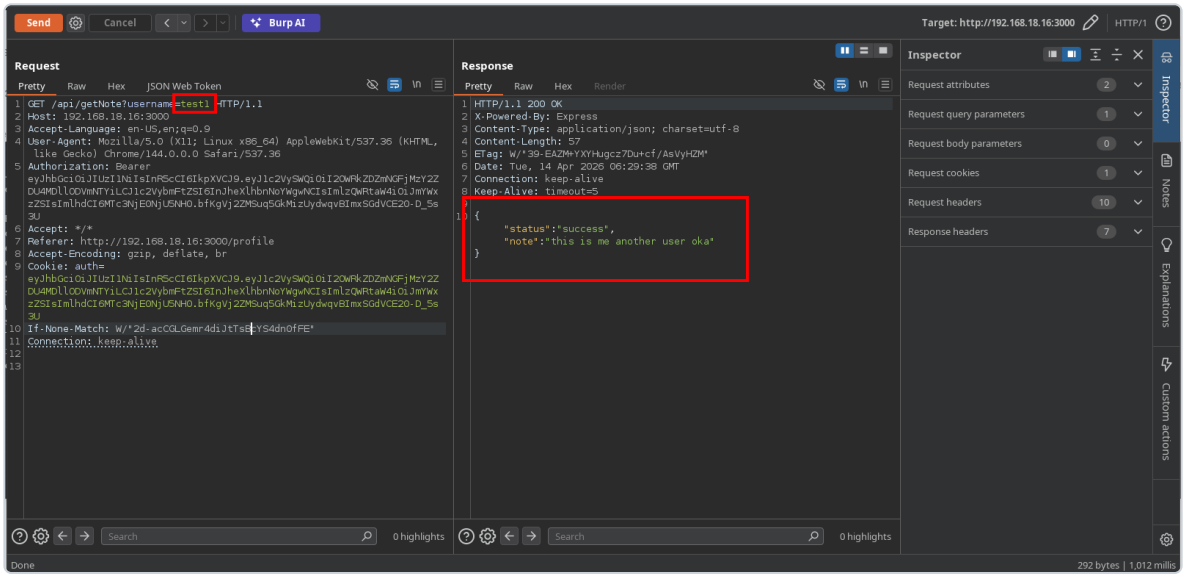
Steps to Reproduce

1. Navigate to the profile page. Intercept traffic in Burp. Identify the request fetching the current user's note: `GET /api/getNote?username=rayyanshah04`.

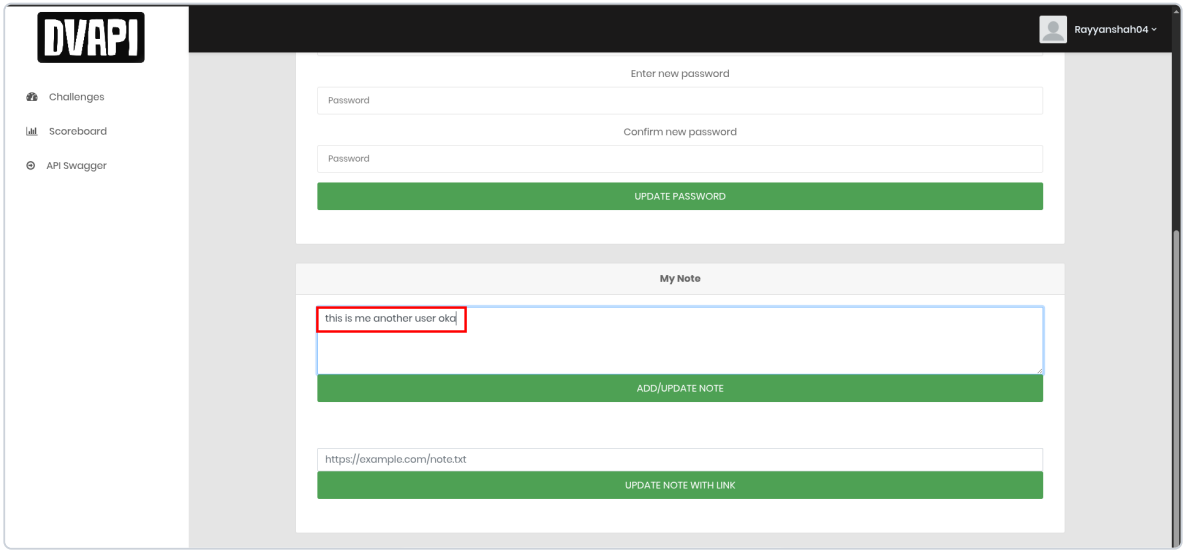


2. Change the `username` parameter to another known user (e.g., `test1`) while keeping the JWT unchanged.

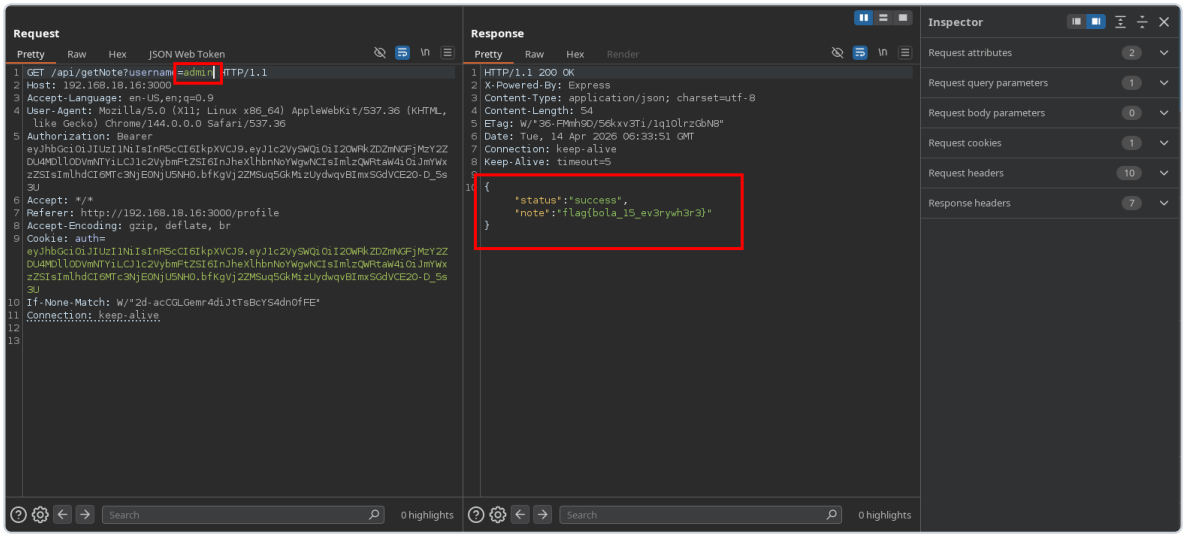
3. Send in Repeater. The server responds `HTTP 200` and returns the target user's private note.



4. The stolen note content renders on the victim's profile page, confirming the data was returned by the server.



5. Repeat with `username=admin`. The admin's private note is returned, which contains the flag.



```
Flag: flag{bola_l5_ev3rywh3r3}
```

Impact

Complete horizontal privilege escalation across all user accounts. Private notes of any user — including admin — are fully accessible to any authenticated attacker.

Remediation

- Derive the target user identity from the authenticated JWT on the server side, never from user-supplied input.
- Implement object-level authorization checks: `if (token.username !== requestedUsername) return 403`.
- Apply this pattern to all resource-fetching endpoints.

FIND-03 — API9:2023 Improper Inventory Management

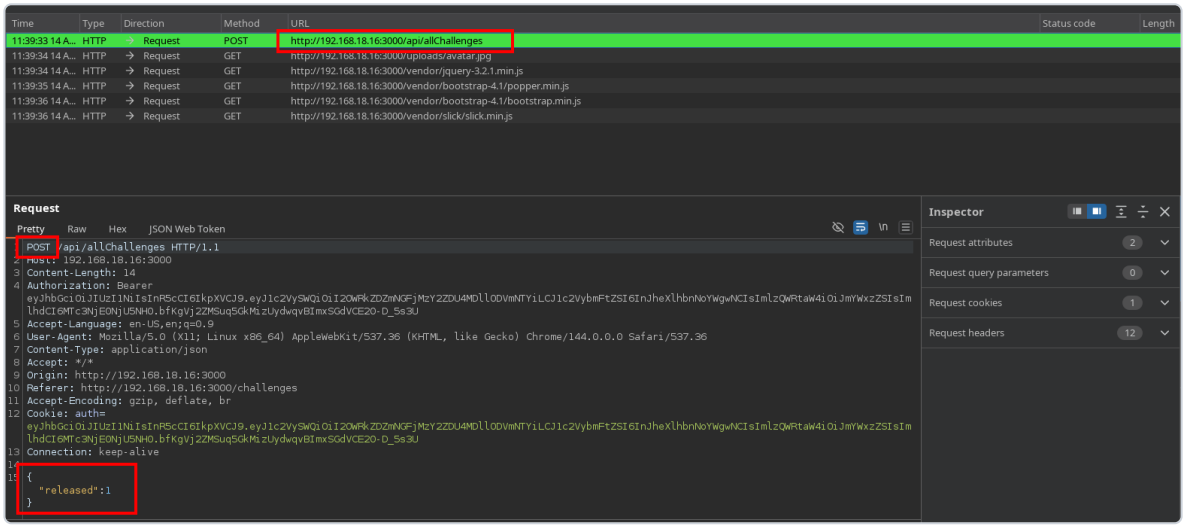
Severity	Medium
Endpoint	`POST /api/allChallenges`
Auth Required	Yes (JWT)
CVSS (approx.)	5.3

Description

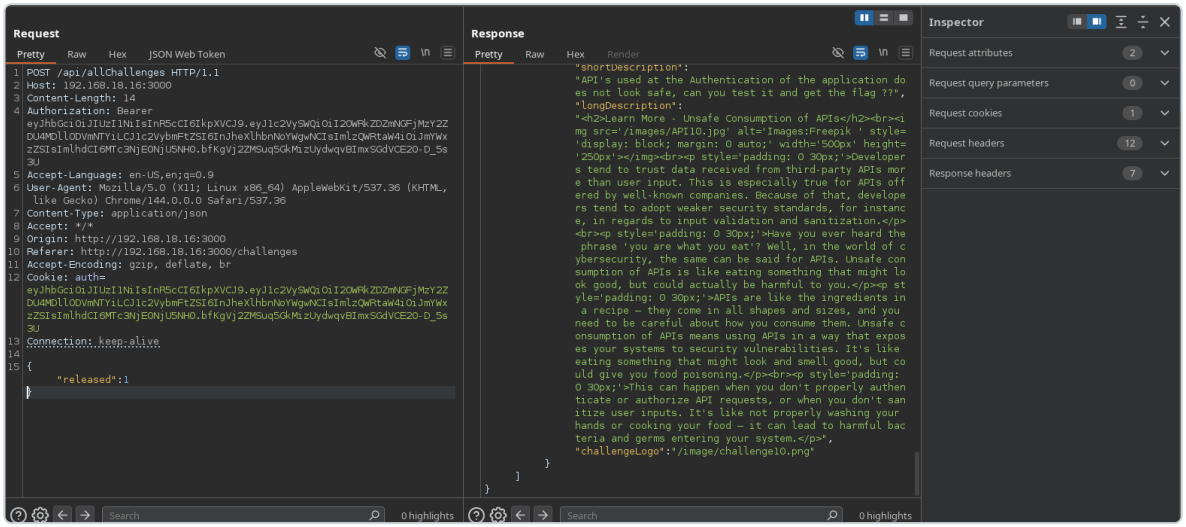
The challenges endpoint uses a POST body parameter `released` to filter which challenges are returned. The server also accepts an undocumented parameter `unreleased` that returns challenges not yet published to users. This exposes hidden application content and internal data through an unretired/shadow API parameter.

Steps to Reproduce

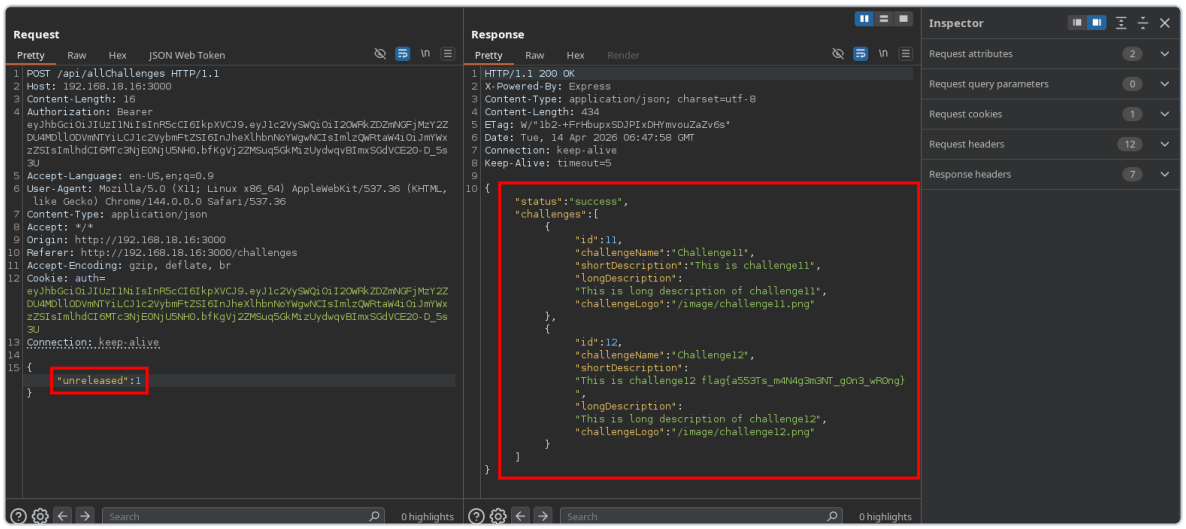
1. Load the challenges page. Intercept in Burp and identify `POST /api/allChallenges` with body `{"released": 1}`.



2. Send to Repeater. Normal response returns 10 challenges as expected.



3. Rename the parameter from `released` to `unreleased` and send. The server returns 2 hidden challenges (Challenge11 and Challenge12), including a flag embedded in the `shortDescription` field.



```
Flag: flag{a553Ts_m4N4g3m3NT_g0n3_wROhg}
```

Impact

Unauthorized access to unreleased features, challenges, and potentially internal API documentation or staging content. Leaks roadmap information and may expose incomplete or insecure features before they are hardened.

Remediation

- Remove or gate all unreleased/shadow parameters server-side. Never rely on obscurity.
- Maintain a versioned API inventory. Decommission or protect endpoints not intended for production use.
- Validate and whitelist accepted input parameters — reject unrecognized keys.

FIND-04 — API3:2023 Broken Object Property Level Authorization

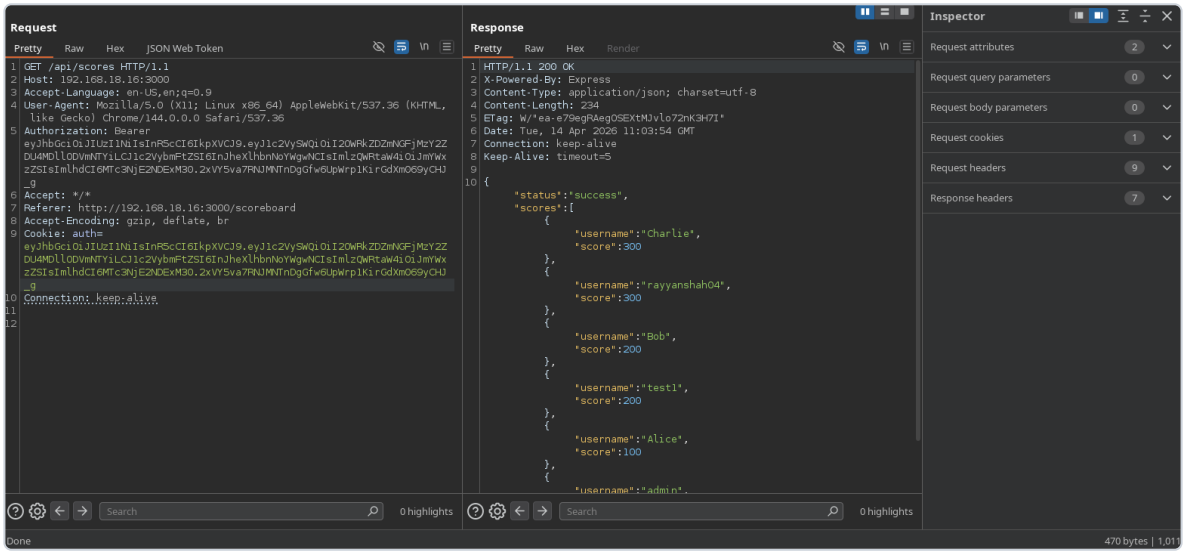
Severity	High
Endpoint	`POST /api/register`
Auth Required	No
CVSS (approx.)	7.5

Description

The user registration endpoint does not restrict which fields a client can supply. By adding a `score` property to the registration request body, an attacker can set an arbitrary score at account creation time. The server blindly persists the injected value into the database without stripping or validating extra properties (mass assignment).

Steps to Reproduce

1. Intercept the scoreboard request to understand the data model: `GET /api/scores` returns `username` and `score` per user.



2. Capture the normal `POST /api/register` request body: `{"username": "apkechaccha", "password": "apkechaccha"}`.

Request

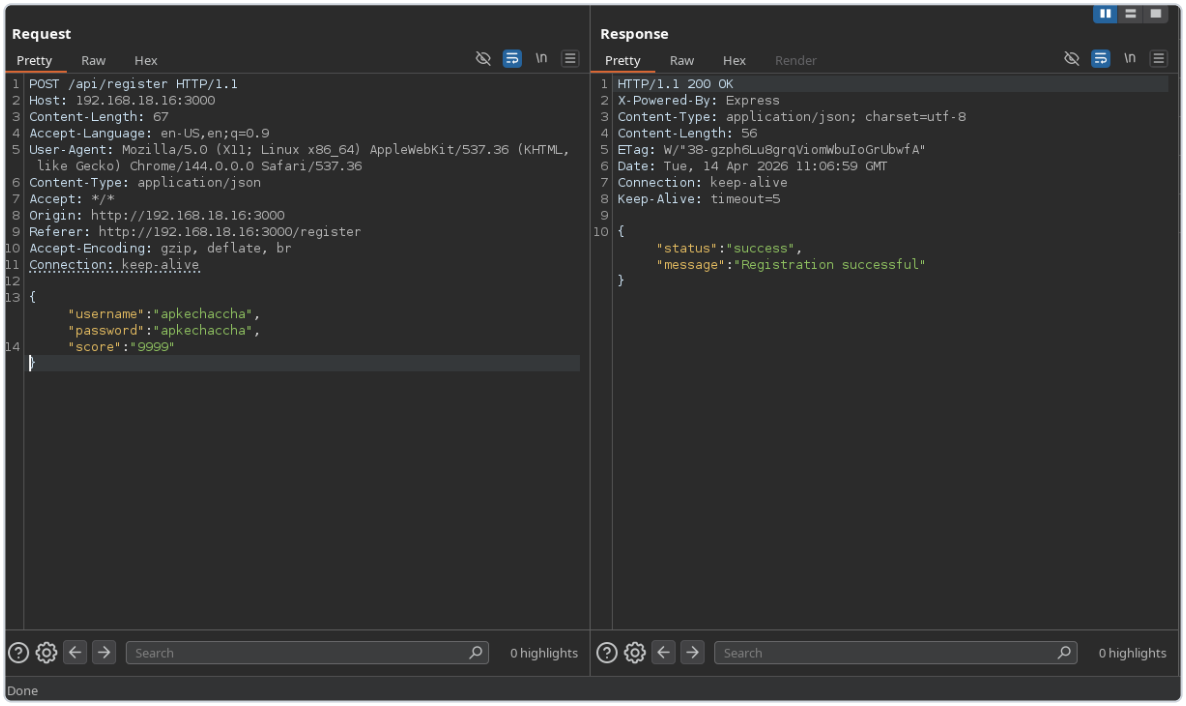
Pretty Raw Hex

```
1  POST /api/register HTTP/1.1
2  Host: 192.168.18.16:3000
3  Content-Length: 67
4  Accept-Language: en-US,en;q=0.9
5  User-Agent: Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/537.36 (KHTML,
    like Gecko) Chrome/144.0.0.0 Safari/537.36
6  Content-Type: application/json
7  Accept: */*
8  Origin: http://192.168.18.16:3000
9  Referer: http://192.168.18.16:3000/register
10 Accept-Encoding: gzip, deflate, br
11 Connection: keep-alive
12
13 {
       "username":"apkechaccha",
       "password":"apkechaccha"
   }
```

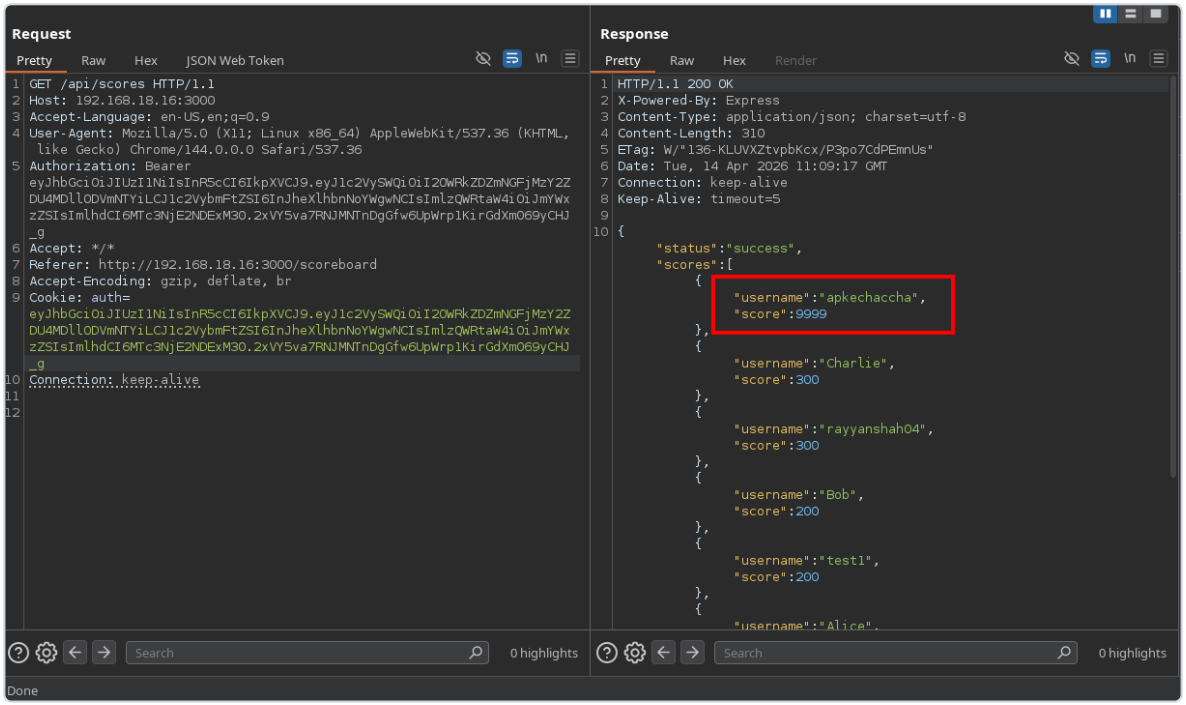
Search 0 highlights

Done

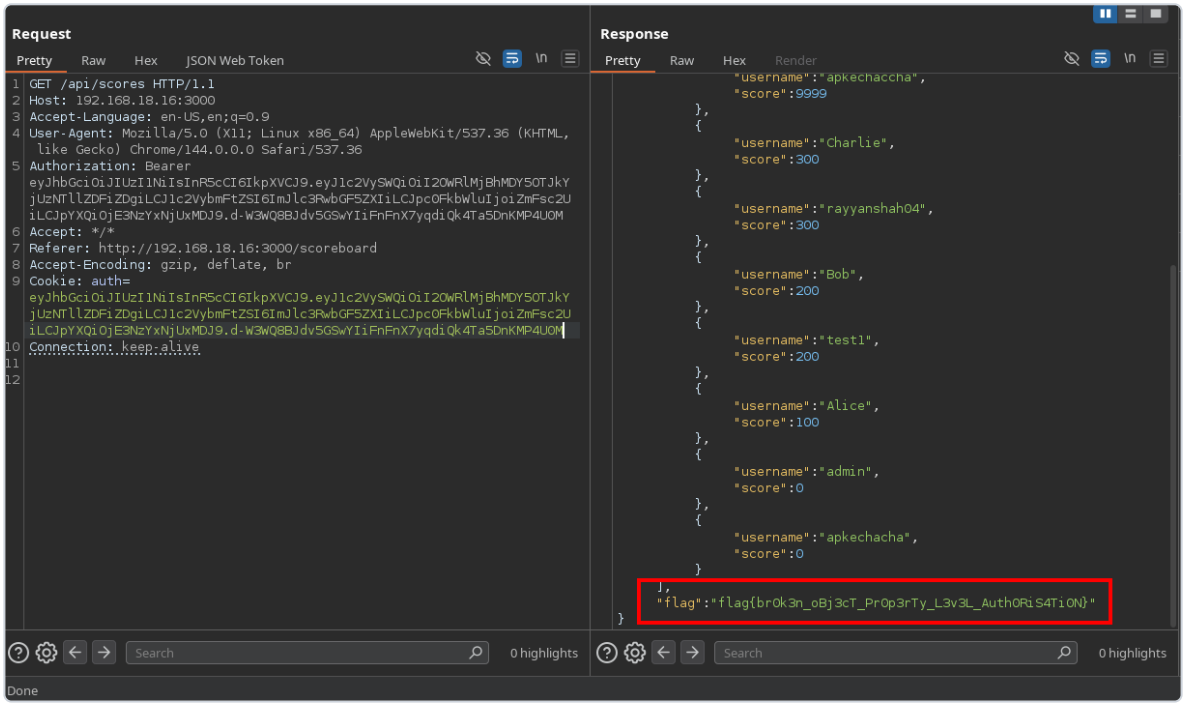
3. Add `"score": "9999"` to the body and resend. The server responds `HTTP 200 – Registration successful`, accepting the injected field.



4. Check `/api/scores`. The new account appears at the top of the scoreboard with score 9999.



5. Scrolling the scores response reveals the flag returned by the API.




```
Flag: flag{brOk3n_oBj3cT_PrOp3rTy_L3v3L_AuthORiS4TiON}
```

Impact

Integrity of all scoreboard data is compromised. Any user can set an arbitrary score at registration, undermining application logic. The same pattern likely applies to other privileged fields (e.g., `isAdmin`, `role`) if they exist in the user schema.

Remediation

- Use an allowlist of accepted fields server-side. Explicitly define and validate the registration schema.
- Strip any unexpected properties before persisting to the database (avoid ORM mass assignment).
- Score should be computed server-side only, never accepted from client input.

FIND-05 — API5:2023 Broken Function Level Authorization (BFLA)

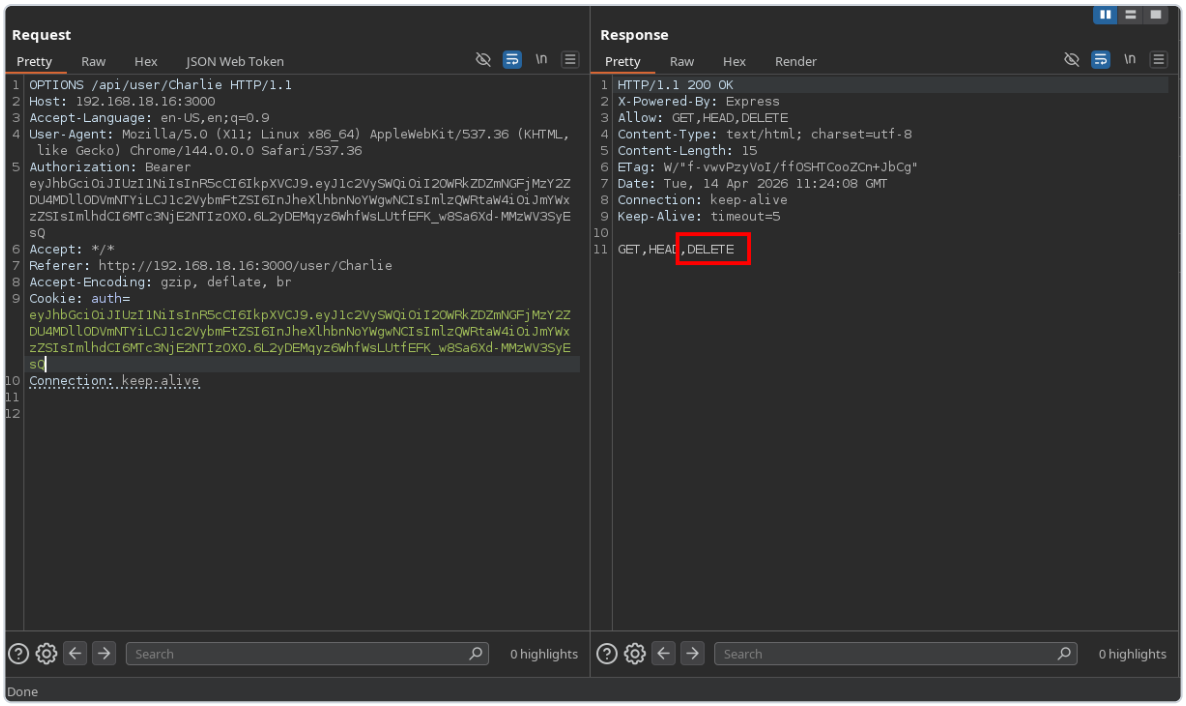
Severity	Critical
Endpoint	`DELETE /api/user/:username`
Auth Required	Yes (any valid JWT)
CVSS (approx.)	9.1

Description

The user profile endpoint exposes a `DELETE` HTTP method that is not accessible through the application UI but is available and functional. Any authenticated user can permanently delete any other user's account by sending a `DELETE` request to `/api/user/<target_username>`. No admin role or ownership check is performed.

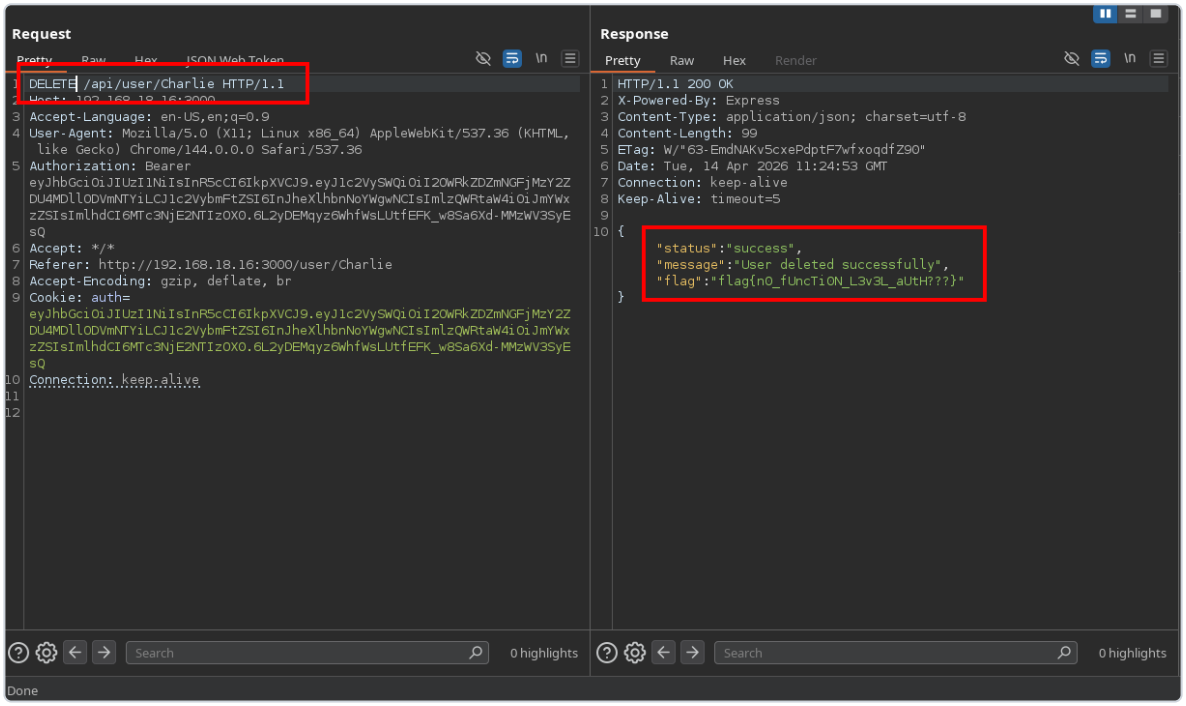
Steps to Reproduce

1. Navigate to a user's profile (e.g., `/user/Charlie`). Capture the request in Burp. Send an `OPTIONS` request to `/api/user/Charlie` to enumerate allowed methods.



   Response header: `Allow: GET, HEAD, DELETE`

2. Send `DELETE /api/user/Charlie` with a standard authenticated JWT (no elevated privileges). The server responds `HTTP 200` and confirms account deletion, returning the flag.




```
Flag: flag{nO_fUncTiON_L3v3L_aUtH???}
```

Impact

Any authenticated user can permanently delete any account on the platform, including admin. This is a complete integrity failure — an attacker can destroy all user data or target specific accounts. No recovery mechanism was observed.

Remediation

- Restrict the `DELETE` method to admin roles only via server-side role checks on the route handler.
- Do not expose administrative HTTP methods on user-facing endpoints. Use separate admin API routes with strict access controls.
- Return `405 Method Not Allowed` for unauthorized methods rather than disclosing available methods via `Allow` headers to non-admin users.

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5. Remediation Summary

ID	Finding	Priority	Fix
FIND-05	Broken Function Level Authorization	Critical	Role-based access control on DELETE routes
FIND-02	Broken Object Level Authorization	High	Derive user identity from JWT server-side
FIND-04	Broken Object Property Level Authorization	High	Allowlist fields in registration schema
FIND-01	Security Misconfiguration (Stack Trace)	High	Disable verbose errors in production
FIND-03	Improper Inventory Management	Medium	Validate and whitelist API parameters

This report was prepared for educational purposes against an intentionally vulnerable application (DVAPI). All findings are within authorized scope.